

Repo Structure

1. ****Pretrain (bootstrap)****

- Script: ``src/alphazero_bot/pretrain.py``
- Purpose: initialize a reasonable policy-value network from historical tensor chunks.
- Output: a starting checkpoint (for example in ``src/models/`` or ``src/alphazero_bot/checkpoints/``).

2. ****Self-play generation****

- Script: ``src/alphazero_bot/self_play.py``
- Purpose: generate training targets from MCTS-guided games using the current ``best`` checkpoint.
- Output: replay files under ``data/selfplay/<iteration_folder>/*.pt`` containing:
 - ``states`` tensor
 - ``policies`` (visit-based target distribution)
 - ``values`` (game outcome from side-to-move perspective)
 - metadata such as ``result`` and ``plies``

3. ****Train candidate****

- Script: ``src/alphazero_bot/train.py``
- Purpose: load self-play replay files, start from ``best`` weights, and optimize a new ``candidate``.
- Output: ``candidate`` checkpoint with updated ``model_state``, optimizer state, and training config.

4. ****Validate (arena)****

- Script: ``src/alphazero_bot/arena.py``
- Purpose: play ``candidate`` vs ``best`` under fixed evaluation settings.
- Output: win/draw/loss score; optional automatic promotion if threshold is met.

5. ****Promote****

- If ``candidate`` beats threshold, copy ``candidate`` -> ``best``.
- If it fails, keep ``best`` unchanged and run the next iteration with better data/settings.

Division of Labor

Preston and Isaiah

Get the data for the chess games. Pre-train two different personalities on the game data.

Simulate games and train the bots using the self-play, train loop to improve bots.

Afomiya and James

Coming up with ways to validate the quality of the chess bots.

Getting chess puzzles, generating reports, comparing performance of different model versions performance, tournament between different bots, and seeing if our bots could solve them as well as playing with our bots online.

GitHub with Readme

https://github.com/pjd9559/Chess_Machine_Learning_Project/tree/codex/alphazero-bot-upload